

Correlation

Association between 2 Numeric Variables

College of Charleston

August, 2026

Today

- ▶ Types of Variables
- ▶ Correlation
- ▶ Graphing?

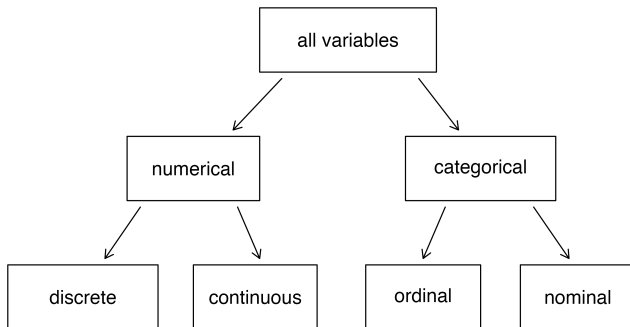
Some definitions

An **observation** (sometimes called an observational unit or case) is the literal thing we are collecting data from and are recording

The information we write down is known as **variables**. Variables typically come in one of two types:

1. **Quantitative Variable:** Typically data that is stored in the form of *numbers*, and is numerical in nature
 - ▶ Continuous data i.e., height and weight
 - ▶ Discrete data (only specific values allowed) i.e., points scored in a game
2. **Categorical Variable:** variables that are naturally divided into *groups*
 - ▶ Nominal (no ordering) ex: eye color
 - ▶ Ordinal (ordering makes sense) ex: year in college (F/J/So/Se)

Variables



Example: Herbicide

A new computer system attempts to spray only weeds (see-and-spray) and not blanket the whole field (full coverage)

Field ID	Spray Type	Wind Speed (mph)	Weed Count	Weeds covered (%)
A	see-and-spray	12	18	72
B	see-and-spray	15	29	55
C	full coverage	8	34	98
D	full coverage	11	18	97
E	see-and-spray	12	24	66
F	full coverage	14	56	95

What is the observational units? What are the variables?

Example: Herbicide

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The different fields (rows) are the obs. units while the variables are the columns (Field ID, Spray Type, Wind Speed, etc...)

The type of variable dictates how we analyze it:

- ▶ Numeric data often uses means (averages)
- ▶ Categorical data often uses proportions (or percentages)
- ▶ Sometimes there are situations in which a variable is technically one type, but it may be more useful to analyze it as another.
 - ▶ Total sales for a grocery store will be recorded in cents but often just say continuous for simplicity
- ▶ Sometimes the type of variable can be different depending on how we record or organize our data.
 - ▶ Age: recorded in years is numeric but young/middle-aged/elderly is categorical

Thoughts on if these might be used as quantitative or categorical variables?

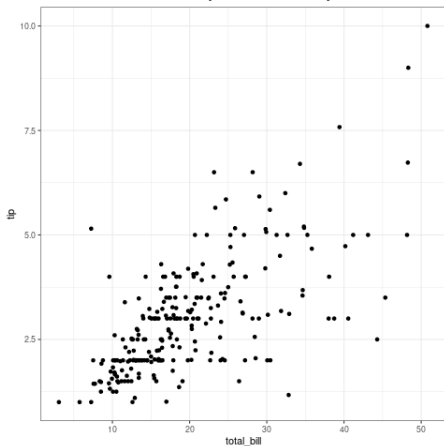
1. Grades for a statistics class
2. A Likert Scale with five levels, measuring pain from "None at all" to "Extreme"
3. The year of birth for people enrolled in MATH 104

"An approximate answer to the right problem is worth a good deal more than an exact answer to an approximate problem."

John Tukey, Statistician

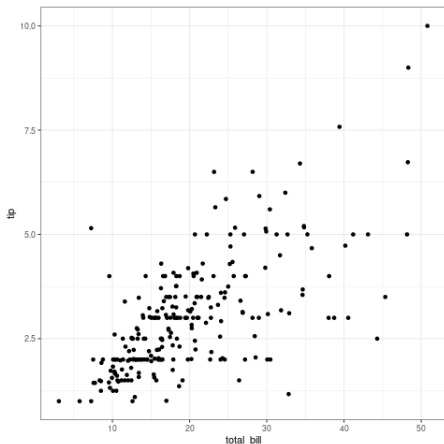
Scatterplots

When we want to plot two quantitative variables → scatterplots



Scatterplots let us see if there are *associations* between quantitative variables

Scatterplots

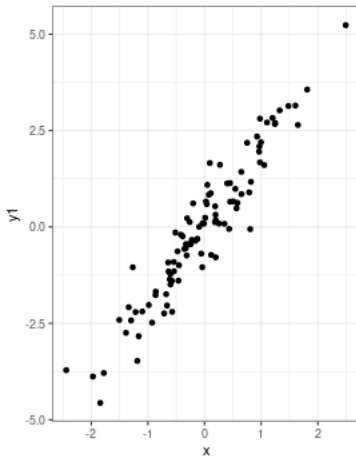
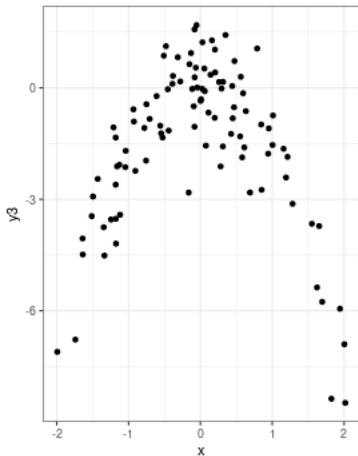


- ▶ The x-axis (horizontal) is the explanatory (independent) variable
- ▶ The y-axis is the response (dependent) variable
- ▶ One point represents one observation
- ▶ Fairly common

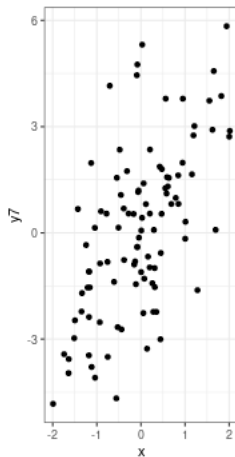
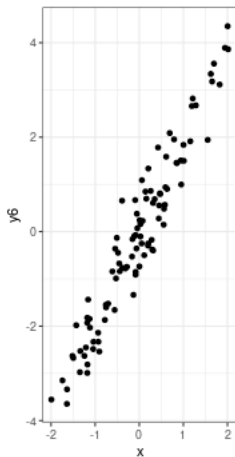
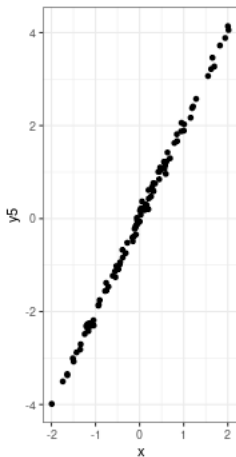
Describing associations in scatterplots:

- ▶ **Form:** pattern? (linear / non-linear / cloud of points)
- ▶ **Strength:** weak / moderate / strong
- ▶ **Direction:** positive / negative
- ▶ **Outliers**

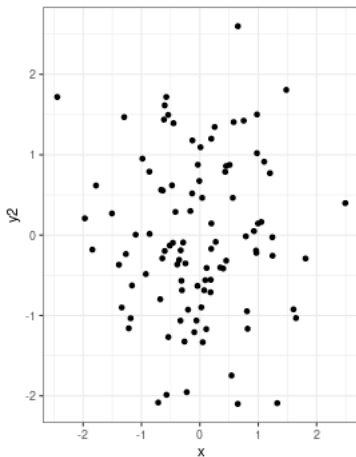
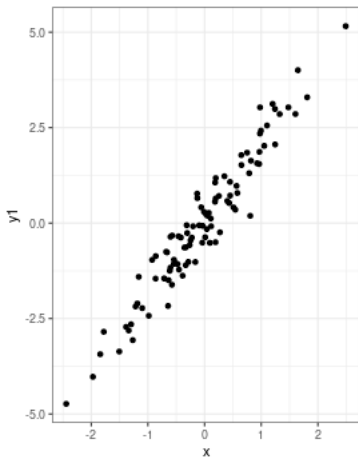
Scatterplot Practice



Scatterplot Practice



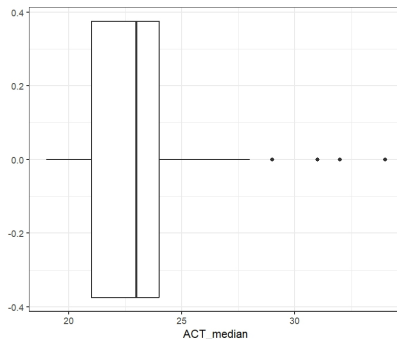
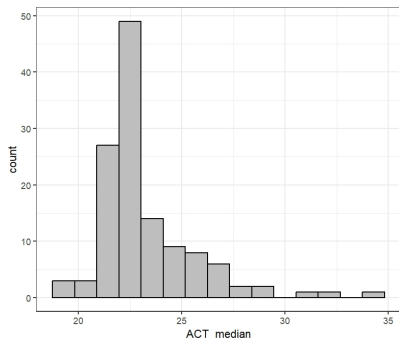
Scatterplot Practice



Extra on Outliers: 1 Numeric Variable

Two ways to look for outliers:

- ▶ histogram → gaps in between the bins (preferred)
- ▶ boxplot → points outside the 'whiskers'

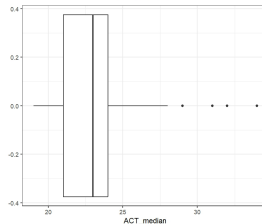
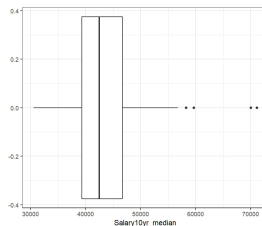
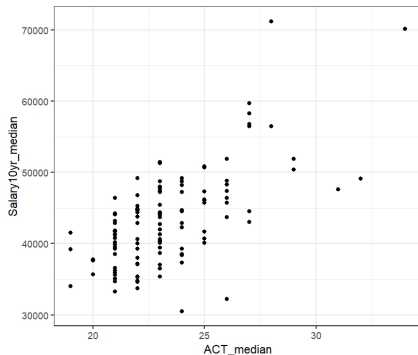


Mention which graph you used!

Extra on Outliers

Outlier in a scatterplot

- ▶ very small or large values for one of the variables (or both!)
- ▶ does not follow the overall pattern



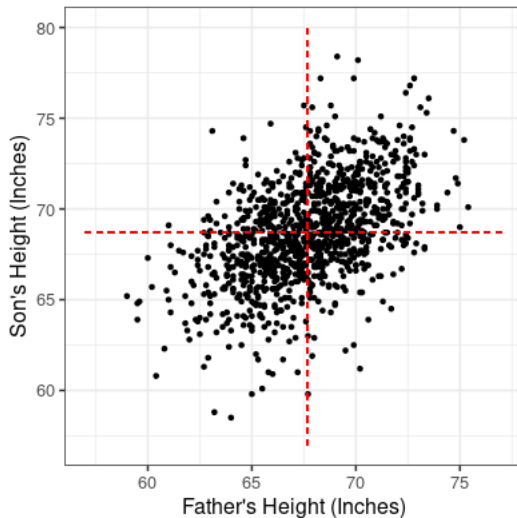
Pearson's Height Data

In the 1880's the Western scientific community was enthralled with the idea of quantifying heritable traits

Karl Pearson collected data on the heights of 1,087 father's and their fully grown first born sons

Father	Son
65.0	59.8
63.3	63.2
65.0	63.3
65.8	62.8
61.1	64.3
63.0	64.2
⋮	⋮

Height Data



Pearson's Correlation Coefficient

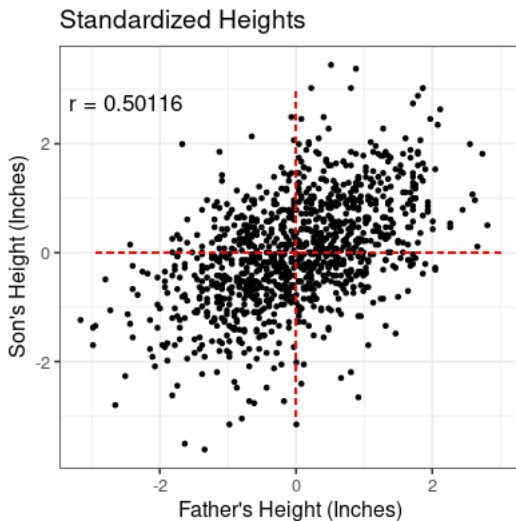
Heights clearly associated, but how to quantify?

Building upon the work from French scientist Francis Galton, Pearson developed the **Pearson's correlation coefficient (r)**:

$$\begin{aligned} r &= \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right) \\ &= \frac{1}{n-1} \sum_{i=1}^n (z_{x_i})(z_{y_i}) \end{aligned}$$

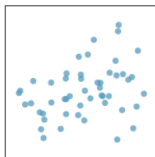
If above-average values of X are common among cases with above-average values of Y (or vice-versa), we should expect r to be positive

Height Data – Standardized

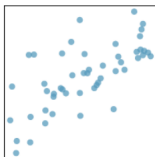


Correlation Examples

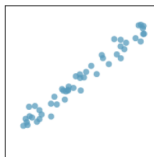
Pearson's correlation coefficient tells us the strength of *linear* association between two quantitative variables (and direction!)



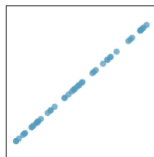
$R = 0.33$



$R = 0.69$



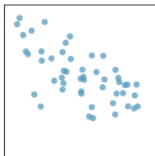
$R = 0.98$



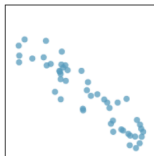
$R = 1.00$



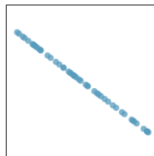
$R = 0.08$



$R = -0.64$



$R = -0.92$



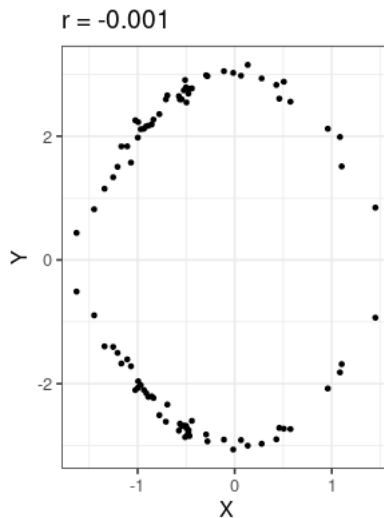
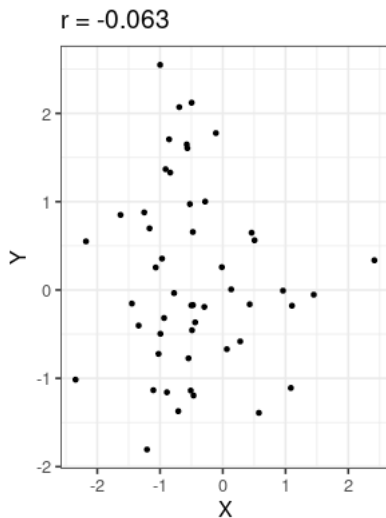
$R = -1.00$

What is considered “strong”?

Correlation Coefficient		Dancey & Reidy (Psychology)	Quinnipiac University (Politics)	Chan YH (Medicine)
+1	-1	Perfect	Perfect	Perfect
+0.9	-0.9	Strong	Very Strong	Very Strong
+0.8	-0.8	Strong	Very Strong	Very Strong
+0.7	-0.7	Strong	Very Strong	Moderate
+0.6	-0.6	Moderate	Strong	Moderate
+0.5	-0.5	Moderate	Strong	Fair
+0.4	-0.4	Moderate	Strong	Fair
+0.3	-0.3	Weak	Moderate	Fair
+0.2	-0.2	Weak	Weak	Poor
+0.1	-0.1	Weak	Negligible	Poor
0	0	Zero	None	None

Source: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6107969/>

Correlation Examples



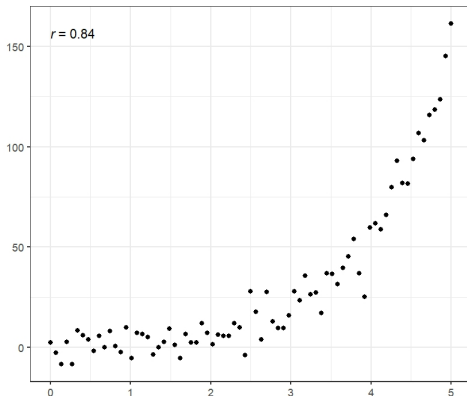
Correlation Properties

Properties:

- ▶ r has no units/is unitless
 - ▶ changing scale of either variable doesn't affect r value
- ▶ r measures the strength of a LINEAR relationship
 - ▶ Gives an idea of how well the scatterplot follows a straight line
 - ▶ Curved lines can be 0, regardless of "strength"
- ▶ r is between -1 and 1
- ▶ The closer r is to 0 $\rightarrow$ weaker linear relationship
- ▶ The closer r is to 1 or -1 $\rightarrow$ stronger linear relationship
- ▶ $r=0 \rightarrow$ no linear relationship

Pitfalls

If we get a value for r close to $+1$ or -1 , it does **not** mean the relationship actually is linear (double-check the scatterplot!)



Pro tip: You should graph before you see the numeric summaries, and as such you should already know there is a curve

Non-linear Association

In addition to Pearson, we have **Spearman's rank correlation** (denoted ρ) where the values of X and Y are replaced with their rank order from smallest to largest before correlating:

$$\begin{array}{lcl} X = \{2, 4, 6, 9, 8\} & \implies & X_{rank} = \{1, 2, 3, 5, 4\} \\ Y = \{7, 4, 1, 5, 3\} & & Y_{rank} = \{5, 3, 1, 4, 2\} \end{array}$$

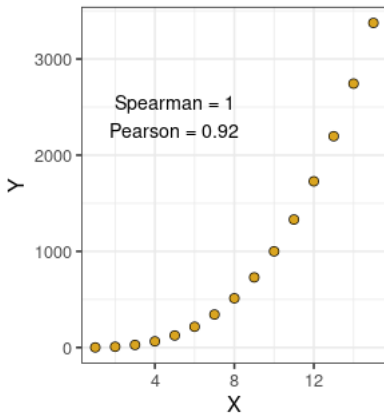
Whereas Pearson's r measures *linear association*, Spearman's ρ measures the *monotonic association* (increasing or decreasing)

- ▶ Can think of this as a non-parameteric version of Pearson's correlation
- ▶ Also can be applied to ordinal data

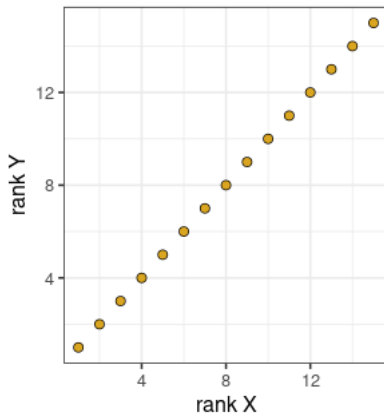
Non-linear Association

$$y = x^3$$

X and Y



Rank X and Rank Y

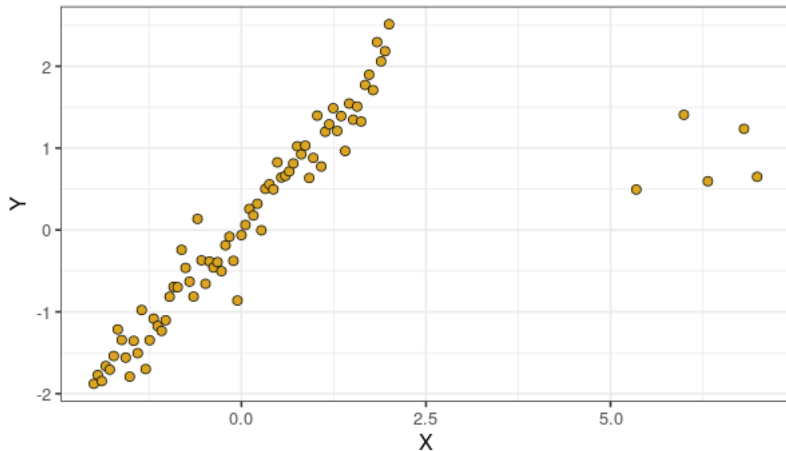


Spearman Correlation

Spearman's correlation is more robust to outliers

Spearman Correlation = 0.95

Pearson Correlation = 0.77



Question

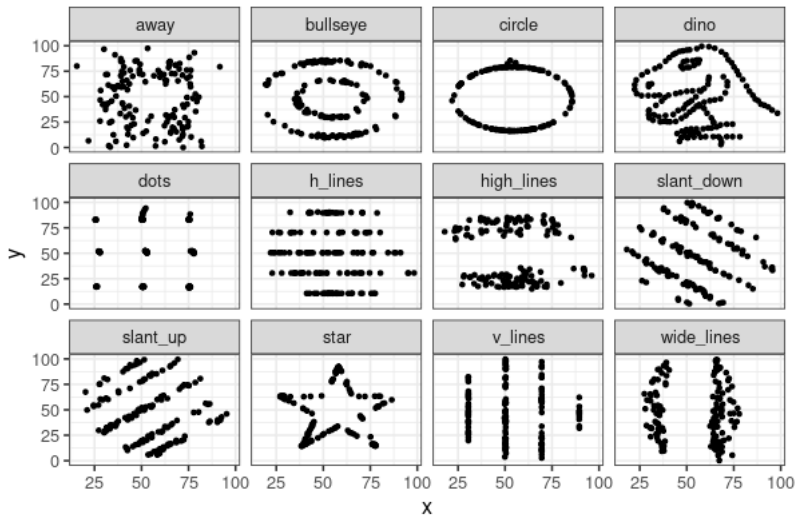
Q: If data sets have the same (or similar) summary statistics do they act the same?

Q: If data sets have the same (or similar) summary statistics do they act/look the same?

No!

- ▶ Summary statistics are summaries only
 - ▶ eg heavily processed data
 - ▶ 7 statistics are an extreme reduction from, say, 400 observations
- ▶ Graphing is the common way to decide that two data sets are similar
- ▶ ?anscombe in R for Anscombe's Quartet data set

"Datasaurus Dozen"



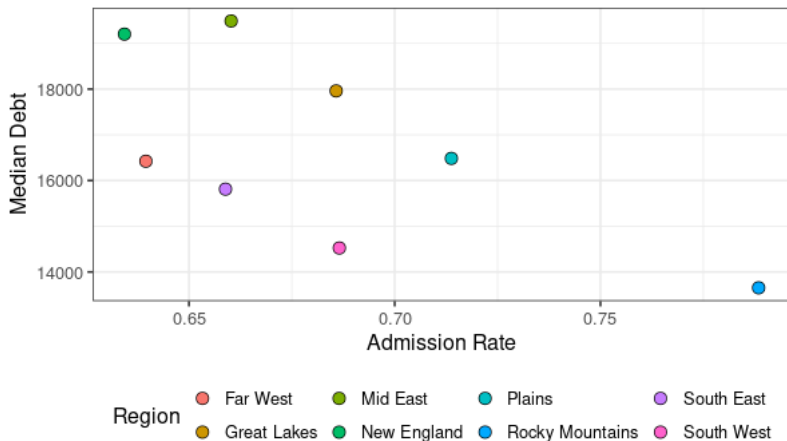
Ecological correlations compare variables for data that have been aggregated at an ecological level

- ▶ Countries
- ▶ States
- ▶ Schools

The *ecological fallacy* is a fallacy in which a conclusion is drawn that, because a correlation exists at a group level, it must exist at the individual level as well

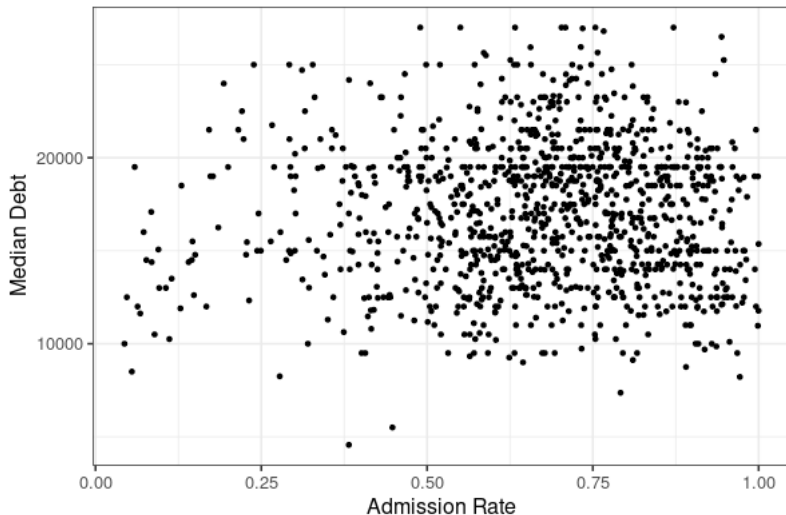
College Ecological Fallacy

Grouping by region, the correlation between (mean) admission rate and (mean) median debt is $r = -0.66$

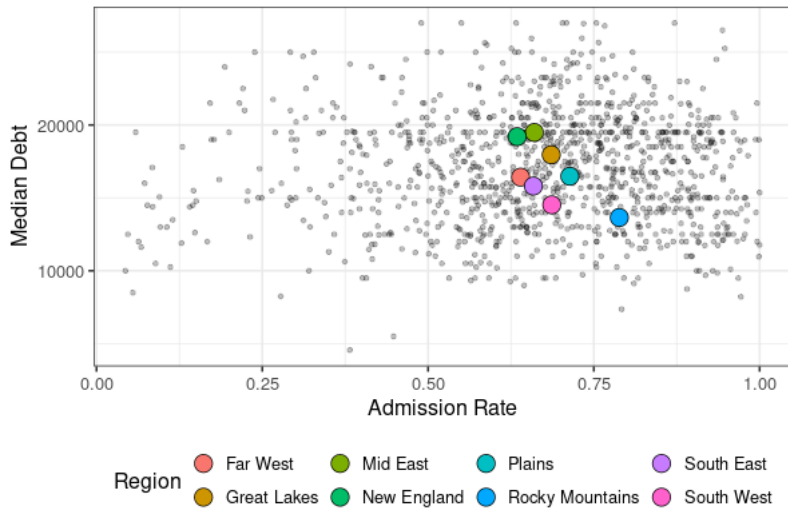


College Ecological Fallacy

This completely disappears when we remove consideration of region, with $r = 0.02$



College Ecological Fallacy



Correlation $\neq$ Causation

We can have a large correlation value between 2 variables. This does not mean the explanatory variable is *causing* a change in the response variable.

Examples with high correlation but where no causal claims can be made:

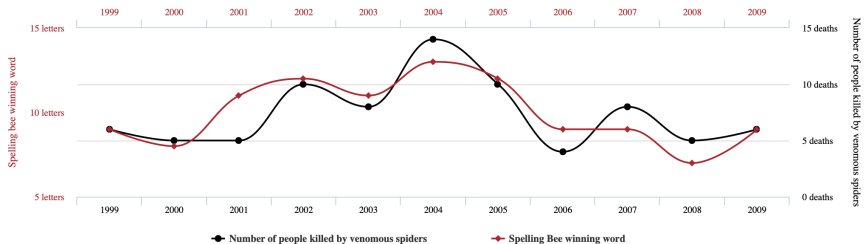
- ▶ Literacy Rate and Gross Domestic Product (GDP) in countries
- ▶ average number of TVs in a household and Life expectancy of countries
- ▶ ice-cream sales and shark attacks

Lurking Variable: a third variable that explains the relationship between two variables with high correlation

Letters in winning word of Scripps National Spelling Bee

correlates with

Number of people killed by venomous spiders



tylervigen.com

- ▶ **Pearson's correlation** strength of *linear association* (and direction)
- ▶ **Spearman rank correlation** useful for data with outlier's or non-linear (but monotone) relationship
- ▶ Be careful with **ecological correlations** – inference for a group is not always valid for individuals
- ▶ Correlation $\neq$ Causation